The 1D Cluster Model as a Symmetry-Protected Topological Phase

This is a reading-only extract of the explanatory cells from the public notebook. For executable code and plots, download the original Jupyter Notebook.

Sanity checks

Three facts underpin everything. Each is one line of code.

1. the stabilizers $S_j = Z_{j-1}X_jZ_{j+1}$ mutually commute, so the model is exactly solvable;
2. the symmetry generators commute with every S_j;
3. on a **ring**, $\prod_{j\ \mathrm{odd}} S_j = U_{\mathrm{odd}}$ — the symmetry is a product of stabilizers, hence acts trivially on the ground state. On an **open chain** this identity fails by boundary terms, and that failure is the whole story.

Fig. 1 — the model

The chain, one stabilizer, the two sublattice symmetries, and the edge operators that will turn out to carry the degeneracy.

Fig. 2 — spectra

An open chain of N sites admits stabilizers S_j only for $j = 2, \dots, N-1$, that is $N-2$ of them, so

$$\dim(\text{ground space}) = \frac{2^N}{2^{N-2}} = 4.$$

This holds for **both parities of** N. On a ring all N stabilizers exist and the ground state is unique. The contrast is the point: the degeneracy is an *edge* phenomenon, not a bulk one.

Fig. 3 — degeneracy versus system size

The ground-space dimension is 4 for every open chain and 1 for every ring, independently of N and of its parity. A degeneracy that does not grow with N and disappears when the boundary is removed is the signature of edge modes.

Fig. 4 — what "protected" means

Add a field and watch the four ground levels.

$$H(\lambda) = -\sum_j Z_{j-1}X_jZ_{j+1} \;-\; \lambda\sum_j P_j, \qquad P = X \text{ or } Z.$$

- $P = X$ **preserves** $\mathbb{Z}_2 \times \mathbb{Z}_2$, since X_j commutes with both $\prod X$.
- $P = Z$ **breaks** it, since Z_j anticommutes with the X on its own sublattice.

The prediction is that the splitting of the four ground levels is exponentially small in N in the first case and linear in λ in the second.

Is it really *exponentially* small?

A splitting that shrinks with system size is a finite-size artefact; one that does not is a genuine lifting of the degeneracy. Fixing λ and increasing N separates the two.

Fig. 5 — entanglement spectrum

Cut the ring in half and record $-\ln \lambda_i$ for the Schmidt weights. At the fixed point the spectrum is exactly flat with four equal levels — one factor of two for each of the two cuts, and each factor of two is the projective representation carried by that cut.

Switching on a perturbation, the spectrum spreads. The rate at which it spreads distinguishes the two cases.

Fig. 6 — the projective representation

On the open chain the symmetry generators, restricted to the ground space, factorise into a left piece and a right piece:

$$U_{\text{odd}}\big|_{\text{GS}} = \underbrace{(X_1 Z_2)}_{L_{\text{odd}}} \cdot \underbrace{(\cdots)}_{R_{\text{odd}}}, \qquad U_{\text{even}}\big|_{\text{GS}} = \underbrace{(Z_1)}_{L_{\text{even}}} \cdot \underbrace{(\cdots)}_{R_{\text{even}}} .$$

The two full operators **commute**, but each edge factor pair **anticommutes**. A $\mathbb{Z}_2 \times \mathbb{Z}_2$ acting with an anticommuting pair has no one-dimensional representation, so each edge must carry at least two states. That is the protection, and $2 \times 2 = 4$ is the degeneracy of Fig. 2.

Fig. 7 — decorated domain walls

Reading the stabilizer for odd j as

$$X_j = Z_{j-1} Z_{j+1}$$

says: the odd spin at j is flipped exactly when the two even neighbours disagree, i.e. **exactly on a domain wall of the even sublattice**. The cluster state is the domain walls of one $\mathbb{Z}_2$ decorated with charges of the other.

This is the construction that generalises: in the "Higgs = SPT" setting the domain walls belong to a higher-form symmetry, but the logic is unchanged.

Summary

observation	figure	meaning
ground space is 4-fold on an open chain, for both parities of N	2, 3	two protected states per edge
ground state is unique on a ring	2	the degeneracy lives on the boundary, not in the bulk
splitting is $\sim 10^{-6}$ under $-\lambda \sum X$, $\sim 10^{-1}$ under $-\lambda \sum Z$	4	the symmetry, not the gap, is what protects
entanglement spectrum is exactly flat and 4-fold at $\lambda = 0$	5	the same projective structure, seen from the cut
the generators commute globally but anticommute on each edge	6	$H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1)) = \mathbb{Z}_2$ is nontrivial
odd-site charges sit on even-sublattice domain walls	7	decorated domain wall construction

Things to try next

- replace $-\lambda \sum Z_j$ by $-\lambda \sum Z_j Z_{j+1}$: it also breaks the symmetry, and the quartet splits again;
- push the symmetric field to $\lambda > 1$: the bulk gap closes and reopens, and the system leaves the SPT phase — the degeneracy is lost through a **bulk** transition rather than an edge one;
- gauge one of the two $\mathbb{Z}_2$'s and watch the SPT turn into a symmetry-breaking phase with two bulk ground states.